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Master's Thesis Summary

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Unraveling and Tuning the Electronic Properties of Topological Quantum Materials

MSc Physics, Quantum Matter | Radboud University Nijmegen

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Topological quantum materials host electronic states associated with nontrivial band topology and provide a platform for investigating unconventional electronic phenomena, with potential applications in spintronics and quantum technologies [1]. Understanding their electronic structure and how it responds to external perturbations is an important topic in condensed matter physics.

The thesis investigates three topological quantum materials under high magnetic fields and at cryogenic temperatures. In the type-II Weyl semimetal WTe_2, the focus is on tuning the Fermi surface through uniaxial strain. In the type-I Weyl semimetal (trigonal) $PtBi_2$, the thesis examines the effects of Pd and Pb doping on its electronic properties and investigates the dimensionality of its superconducting state. Finally, the intrinsic magnetic topological insulator $MnSb_2Te_4$ is studied to characterize its magnetotransport properties and search for quantum oscillations associated with its Fermi surface.

Experiments were conducted at the High Field Magnet Laboratory (HFML) in Nijmegen using superconducting magnets and helium bath cryostats, reaching temperatures down to approximately 300 mK. Electronic transport measurements were performed in a Hall-bar geometry using AC excitation and lock-in detection to record longitudinal and transverse voltages. In $PtBi_2$, cantilever torque magnetometry complemented transport measurements by providing a more sensitive probe of quantum oscillations. Measurements were performed as a function of magnetic field and temperature for different field orientations to investigate the electronic properties of the materials. Sample preparation was carried out by hand, from cutting single crystals and attaching gold-wire electrical contacts to mounting the samples on the measurement probes. For WTe_2, the samples were glued to a novel flexure mounted on a piezoelectric strain cell, enabling the application of uniaxial strain, which was quantified through capacitance measurements. The strain-cell setup did not allow angle-dependent measurements.

WTe_2 exhibited an extremely large, non-saturating quadratic magnetoresistance up to 16 T, consistent with nearly compensated electron and hole carriers [2]. The magnetoresistance was modified by applied uniaxial strain, while Shubnikov–de Haas oscillations revealed corresponding changes in the Fermi surface frequencies under both compressive and, for the first time, tensile strain. These results demonstrate that uniaxial strain can tune the Fermi surface of WTe_2. The observed frequency shifts were larger than those reported in previous studies, and their dependence on compressive strain differed from the previously reported linear behavior [3]. These discrepancies may reflect differences in strain application methods, although further measurements are needed to establish their origin and assess the reproducibility of the response.

Pristine **PtBi**$_2$ exhibited a large anisotropic magnetoresistance superimposed with quantum oscillations. Light Pd and Pb substitution reduced the magnetoresistance by approximately an order of magnitude, while no quantum oscillations were detectable in the doped samples by transport measurements. These changes are consistent with increased carrier scattering due to doping-induced disorder. Cantilever torque magnetometry revealed nearly identical quantum oscillation frequencies in pristine and Pb-doped $PtBi_2$, suggesting that Pb substitution does not substantially change the Fermi surface. The measurements also resolved additional oscillation frequencies not reported in previous studies, revealing further details of the quantum oscillation spectrum [4]. These frequencies may originate from previously unidentified Fermi pockets or composite orbits, although further investigation is needed to determine their origin. Separately, the superconducting properties of pristine $PtBi_2$ were investigated through electronic transport. The temperature dependence and angular anisotropy of the superconducting critical field provided evidence for a two-dimensional superconducting state. The observed two-dimensional superconducting behavior is particularly relevant given reports of proposed topological surface superconductivity in $PtBi_2$ [5, 6, 7]. These results contribute to the broader effort to establish the nature of superconductivity in $PtBi_2$, which requires complementary experimental techniques to distinguish between surface and bulk contributions.

MnSb$_2$**Te**$_4$ exhibited anisotropic, non-monotonic magnetoresistance that initially decreased and then increased with magnetic field. No quantum oscillations were observed in the transport measurements. The temperature at which the magnetoresistance reached its minimum was close to the magnetic transition temperature, which was also reflected in a peak in the zero-field resistance. This behavior suggests competing contributions from field-induced spin alignment and orbital effects [8]. A further resistance upturn below 3 K was observed, but its origin remains unresolved. It may involve contributions from topological surface states, although additional measurements are needed to distinguish this possibility from alternative mechanisms.

References

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